

2. If we receive a measurement $y(k_0)$, where $k_0 < k$, then we can update the state estimate and its covariance to $\hat{x}(k, k_0)$ and $P(k, k_0)$ as follows.

- (a) Retrodict the state estimate from k back to k_0 as shown in Equation (10.118):

$$\begin{aligned} S(k) &= H(k)P^-(k)H^T(k) + R(k) \\ \hat{x}(k_0, k) &= F(k_0, k) [\hat{x}(k) - Q(k, k_0)H^T(k)S^{-1}(k)r(k)] \end{aligned} \quad (10.128)$$

- (b) Compute the covariance of the retrodicted state using Equations (10.122) and (10.123):

$$\begin{aligned} P_w(k, k_0) &= Q(k, k_0) - Q(k, k_0)H^T(k)S^{-1}(k)H(k)Q(k, k_0) \\ P_{xw}(k, k_0) &= Q(k, k_0) - P^-(k)H^T(k)S^{-1}(k)H(k)Q(k, k_0) \\ P(k_0, k) &= F(k_0, k) \{ P(k) - P_{xw}(k, k_0) - P_{xw}^T(k, k_0) + \\ &\quad P_w(k, k_0) \} F^T(k_0, k) \end{aligned} \quad (10.129)$$

- (c) Compute the covariance of the retrodicted measurement at time k_0 using Equation (10.124):

$$S(k_0) = H(k_0)P(k_0, k)H^T(k_0) + R(k_0) \quad (10.130)$$

- (d) Compute the covariance of the state at time k and the retrodicted measurement at time k_0 using Equation (10.125):

$$P_{xy}(k, k_0) = [P(k) - P_{xw}(k, k_0)]F^T(k_0, k)H^T(k_0) \quad (10.131)$$

- (e) Use the delayed measurement $y(k_0)$ to update the state estimate and its covariance:

$$\begin{aligned} \hat{x}(k, k_0) &= \hat{x}(k) + P_{xy}(k, k_0)S^{-1}(k_0)[y(k_0) - H(k_0)\hat{x}(k_0, k)] \\ P(k, k_0) &= P(k) - P_{xy}(k, k_0)S^{-1}(k_0)P_{xy}^T(k, k_0) \end{aligned} \quad (10.132)$$

It is possible to make some simplifying approximations to this delayed measurement filter in order to decrease computational cost with only a slight degradation in performance [Bar02].

10.6 SUMMARY

In this chapter we discussed some important topics related to Kalman filtering that extend beyond standard results. We have seen how to verify if a Kalman filter is operating reliably. This gives us a quantifiable confidence in the accuracy of our filter estimates. We also discussed multiple-model estimation, which is a way of estimating system states when we are not sure of which model is governing the dynamics of the system. This can be useful when the system model changes in unpredictable ways. We discussed reduced-order filtering, which can be used to estimate a subset of the system states while saving computational effort. We derived a robust Kalman filter, which makes the filter less sensitive to variations

in the assumed system model. Robust filtering naturally leads into the topic of H_∞ filtering, which we will discuss in Chapter 11. Finally, we derived a way to update the state estimate when a measurement arrives at the filter in the wrong chronological order because of processing delays.

There are several other important extensions to Kalman filtering that we have not had time to discuss in this chapter. One is the variable structure filter, which is a combination of the Kalman filter with variable structure control. This guarantees stability under certain conditions and often provides performance better than the Kalman filter, especially when applied to nonlinear systems [Hab03]. Another recent proposal is the proportional integral Kalman filter, which adds an integral term to the measurement state update and thereby improves stability and reduces steady-state tracking errors [Bas99]. Another interesting topic is the use of a perturbation estimator to estimate the process noise. This allows model uncertainties to be lumped with process noise so that the process-noise estimate increases the robustness of the filter [Kwo03].

PROBLEMS

Written exercises

10.1 In this problem we consider the scalar system

$$\begin{aligned}x_{k+1} &= x_k + w_k \\ y_k &= x_k + v_k\end{aligned}$$

where w_k and v_k are white and uncorrelated with respective variances Q and R , which are unknown. A suboptimal steady-state value of K is used in the state estimator since Q and R are unknown.

- Use the expression for P_k^- along with the first expression for P_k^+ in Equation (5.19) to find the steady-state value of P_k^- as a function of the suboptimal value of K and the true values of Q and R . [Note that the first expression for P_k^+ in Equation (5.19) does not depend on the value for K_k being optimal.]
- Now suppose that $E(r_k^2)$ and $E(r_{k+1}r_k)$ are found numerically as the filter runs. Find the true value of R and the steady-state value of P_k^- as a function of $E(r_k^2)$ and $E(r_{k+1}r_k)$.
- Use your results from parts (a) and (b) to find the true value of Q .

10.2 Show that the innovations $r = y - C\hat{x}$ of the continuous-time Kalman filter is white with covariance R .

10.3 Consider the system described in Problem 5.1. Find the steady-state variance of the Kalman filter innovations when $Q = R$ and when $Q = 2R$.

10.4 Consider the system of Problem 10.3 with $Q = R = 1$. Suppose the Kalman filter for the system has reached steady state. At time k the innovations $r_k = y_k - \hat{x}_k^-$.

- Find an approximate value for $\text{pdf}(y_k|p)$ (where p is the model used in the Kalman filter) if $r_k = 0$, if $r_k = 1$, and if $r_k = 2$.

- b) Suppose that the use of model p_1 gives $r_k = 0$, model p_2 gives $r_k = 1$, and model p_3 gives $r_k = 2$. Further suppose that $\Pr(p_1|y_{k-1}) = 1/4$, $\Pr(p_2|y_{k-1}) = 1/4$, and $\Pr(p_3|y_{k-1}) = 1/2$. Find $\Pr(p_j|y_k)$ for $j = 1, 2, 3$.

10.5 Consider the system described in Example 4.1 where the measurement consists of the predator population. Suppose that we want to estimate $x(1) + x(2)$, the sum of the predator and prey populations. Create an equivalent system with transformed states such that our goal is to estimate the first element of the transformed state vector.

10.6 Consider the system

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w_k \\ y_k &= \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + v_k \end{aligned}$$

where w_k and v_k are uncorrelated zero-mean white noise processes with variances q and R , respectively.

- Use Anderson's approach to reduced-order filtering to estimate the first element of the state vector. Find steady-state values for $\tilde{P}$, $\tilde{\Sigma}$, $\tilde{\Pi}$, and P . Find the steady-state gain K of the reduced-order filter.
- Use the full-order filter to estimate the entire state vector. Find steady-state values for P and K .
- Comment on the comparison between your answer for P in part (a) and part (b).

10.7 Consider the reduced-order filter of Example 10.3 with the initial condition $\tilde{P}_0^+ = 1$.

- Find analytical expressions for the steady-state values of $\tilde{K}$, α , $\tilde{P}^+$, Σ^+ , $\tilde{P}^-$, $\tilde{P}^+$, $\tilde{P}^-$, Σ^- , and $\tilde{P}^-$.
- What does the reduced-order filter indicate for the steady-state *a posteriori* estimation-error variance of the first state? Find an analytical expression for the true steady-state *a posteriori* estimation-error variance of the first state when the reduced-order filter is used. Your answer should be a function of $x(2)$. Solve for the true steady-state *a posteriori* estimation-error variance of the first state when $x(2) = 0$, when $x(2) = 1$, and when $x(2) = 2$.
- What is the steady-state *a posteriori* estimation-error variance of the first state when the full-order filter is used?

10.8 Verify that the two expressions in Equation (10.98) are respectively equal to the cross-covariance of x and y , and the covariance of y .

10.9 Suppose you have the linear system $x_{k+1} = Fx_k + w_k$, where $w_k \sim (0, Q_k)$ is zero-mean white noise. Define $w(k+2, k)$ as the cumulative effect of all of the process noise on the state from time k to time $(k+2)$. What are the mean and covariance of $w(k+2, k)$?

10.10 Suppose that a Kalman filter is running with

$$F = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
H &= \begin{bmatrix} 1 & 0 \end{bmatrix} \\
Q &= \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\
R &= 1 \\
P^+(k) &= \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix}
\end{aligned}$$

An out-of-sequence measurement from time $(k-1)$ is received at the filter.

- What was the value of $P^-(k)$?
- Use the delayed-measurement filter to find the quantities $P_w(k, k-1)$, $P_{xw}(k, k-1)$, $P(k-1, k)$, $P_{xy}(k, k-1)$, and $P(k, k-1)$.
- Realizing that the measurement at time $(k-1)$ was not received at time $(k-1)$, derive the value of $P^-(k-1)$. Now suppose that the measurement was received in the correct sequence at time $(k-1)$. Use the standard Kalman filter equations to compute $P^+(k-1)$, $P^-(k)$, and $P^+(k)$. How does your computed value of $P^+(k)$ compare with the value of $P(k, k-1)$ that you computed in part (b) of this problem?

10.11 Under what conditions will P_y in Equation (10.100) be invertible for all k ?

Computer exercises

10.12 Consider the equations

$$\begin{aligned}
300x + 400y &= 700 \\
100x + 133y &= 233
\end{aligned}$$

- What is the solution of these equations?
- What is the solution of these equations if each constant in the second equation increases by 1?
- What is the condition number of the original set of equations?

10.13 Repeat Problem 10.12 for the equations

$$\begin{aligned}
300x + 400y &= 700 \\
100x + 200y &= 200
\end{aligned}$$

Comment on the difference between this set of equations and the set given in Problem 10.12.

10.14 Tire tread is measured every τ weeks. After τ weeks, 20% of the tread has worn off, so we can model the dynamics of the tread height as $x_{k+1} = fx_k + w_k$, where $f = 0.8$, and w_k is zero-mean white noise with a variance of 0.01. We measure the tread height every τ weeks with zero-mean white measurement noise that has a variance of 0.01. The initial tread height is known to be exactly 1 cm. Write a program to simulate the system and a Kalman filter to estimate the tread height.

- Run the program for 10 time steps per tire, and for 1000 tires. What is the mean of the 10,000 measurement residuals?

- b) Suppose the Kalman filter designer incorrectly believes that 30% of the tread wears off every τ weeks. What is the mean of the 10,000 measurement residuals in this case?
- c) Suppose the Kalman filter designer incorrectly believes that 10% of the tread wears off every τ weeks. What is the mean of the 10,000 measurement residuals in this case?

10.15 Consider the system described in Problem 10.14. Suppose the engineer does not know the true value of f but knows the initial probabilities $\Pr(f = 0.8) = \Pr(f = 0.85) = \Pr(f = 0.9) = 1/3$. Run the multiple-model estimator for 10 time steps on 100 tires to estimate f . The f probabilities at each time step can be taken as the mean of the 100 f probabilities that are obtained from the 100 tire simulations, and similarly for the f estimate at each time step. Plot the f probabilities and the f estimate as a function of time.

10.16 Consider a scalar system with $F = H = 1$ and nominal noise variances $Q = R = 5$. The true but unknown noise variances $\tilde{Q}$ and $\tilde{R}$ are given as

$$\begin{aligned}\tilde{Q} &= (1 + \alpha)Q \\ \tilde{R} &= (1 + \beta)R \\ E(\alpha^2) &= \sigma_1^2 = 1/2 \\ E(\beta^2) &= \sigma_2^2 = 1\end{aligned}$$

where α and β are independent zero-mean random variables. The variance of the *a posteriori* estimation error is P if $\alpha = \beta = 0$. In general, α and β are nonzero and the variance of the estimation error is $P + \Delta P$. Plot P , $E(\Delta P^2)$, and $(P + E(\Delta P^2))$ as a function of K for $K \in [0.3, 0.7]$. What are the minimizing values of K for the three plots?

PART III

THE H_∞ FILTER
